

Tasrifur Riahi

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■ EDUCATION

Degree	Institution	Grade	Year
M.Sc. Engineering in Information and Communication Technology (ICT)	Bangladesh University of Engineering and Technology (BUET), Dhaka	CGPA 3.92/4.00	2024 – 2026 (exp.)
B.Sc. (Engineering) in Electronics and Communication Engineering	Hajee Mohammad Danesh Science and Technology University (HSTU), Dinajpur	CGPA 3.64/4.00, 3rd Position	2018 – 2023
HSC & SSC (Science)	Jhitka Khawja Rahmat Ali Degree College & Jhitka Ananda Mohan High School	GPA 4.83 & 5.00 (of 5.00)	2017 & 2015

■ RESEARCH INTEREST

Computer Vision, Gait Analysis & Recognition, Machine Learning, Deep Learning, Neural Networks, Pattern Recognition, Image & Video Processing, Internet of Things (IoT).

■ RESEARCH EXPERIENCE

Postgraduate Research Fellow

June 2025 – Present

Information Security Lab, IICT, BUET, Dhaka | Supervisor: Prof. Dr. Hossen Asiful Mustafa

- Developing a few-shot learning method for gait recognition that stays robust under appearance variations (clothing, carried objects, viewpoint).
- Poster on this work placed **1st Runner Up** at the IICT Student Poster Competition 2026, BUET.

Undergraduate Thesis Student

2022 – 2023

Dept. of Electronics and Communication Engineering, HSTU, Dinajpur | Supervisor: Prof. Md. Mehedi Islam

- Thesis:** *ElderFallGuard* — a real-time, non-invasive elderly fall-detection system using MediaPipe pose estimation and machine-learning classifiers.
- Built a custom 7,200-sample dataset across 12 poses; Random Forest achieved the best performance, with instant Telegram caregiver alerts. Published as a first-author arXiv preprint.

■ ACADEMIC PROJECTS

Automatic Fire Fighter Robot.

- Autonomous flame detection and suppression prototype; Second Runner-Up at the Robotics & Mechatronics Project Hub.

Calorie Burn Prediction from Step-Counter Data.

- Time-series machine-learning model estimating energy expenditure from step counts.

Hospital Management System

- Object-oriented software project.

■ SCHOLARLY CONTRIBUTIONS

Preprint

- **T. Riahi**, M. A. H. Bappy, M. M. Islam, “ElderFallGuard: Real-Time IoT and Computer Vision-Based Fall Detection System for Elderly Safety,” *arXiv:2505.11845*, 2025. [link]

Poster

- **T. Riahi**, “A Few-Shot Learning Method in Gait Recognition under Appearance Variations,” IICT Student Poster Competition 2026, BUET, Dhaka — **2nd place**, Apr. 2026.

■ RESEARCH SKILLS

- **Areas of Expertise:** Computer vision, machine learning, deep learning, pattern recognition, IoT systems, data analysis, dataset design.
- **Software & Tools:** MATLAB, MediaPipe, scikit-learn, Proteus, PSpice, Packet Tracer, MS Office.
- **Programming Languages:** Python, C, PHP, JavaScript, SQL (MySQL).

■ ACHIEVEMENTS & SCHOLARSHIPS

- **Excellence in Event Management**, Shahid Smriti Hall, BUET (Aug 2026).
- **First Runner Up**, IICT Student Poster Competition 2026, BUET (gait recognition).
- **Dean’s List Award (twice)**, B.Sc. (Engineering) in (ECE), HSTU.
- **Second Runner-Up**, Robotics & Mechatronics Project Hub.

■ CERTIFICATIONS & WORKSHOPS

- Asia Pacific Internet Engineering Local Camp — IICT, BUET & SOI Asia.
- Internship — Bangladesh Telecommunications Company Limited (BTCL).
- Data Science Certification Program — Ostad.
- Web Design and development — LEDP, ICT Division, Bangladesh
- Basics of Web Application & Postman — Ostad.
- Frontend Fundamentals — Pirple.
- Training on Technologies for the Fourth Industrial Revolution — ICT Division, Bangladesh.

■ EXTRA-CURRICULAR ACTIVITIES

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| • Vice President, ECE Club of HSTU | Feb 2023 – Dec 2023 |
| • Campus Ambassador, GUVI (IIT-Madras incubated) | Sep 2022 – Feb 2023 |
| • Campus Ambassador, Ostad | Jan 2022 – Jan 2023 |
| • Campus Ambassador, Bangladesh–China Engineering Forum | 2020 – 2021 |

■ REFERENCES

Dr. Hossen Asiful Mustafa

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Institute of Information and Communication
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